Data Sheet

EXPERIMENT NO. 08

Title: - Simulation of Summing Amplifier using Op-Amp (using Tinkercad / Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity
1	OP-Amp	IC 741	1
2	Resistor	10KΩ, 20KΩ, 47KΩ	3
3	Multimeter	Digital Display	1
4	Power Supply	15V	3
5	Bread board	-	1
6	Connecting Wires	-	required

Important Formulae: -

Inverting Summing Amplifier (general case): -

$Vo = -R_f \times [(V1/R1) + (V2/R2) + (V3/R3)]$

- V1, V2, V3 = Input voltages
- R1, R2, R3 = Input resistors
- R_f = Feedback resistor

Equal Input Resistors (R1 = R2 = R3 = R): -

$Vo = -(R_f/R) \times (V1 + V2 + V3)$

Unity-Gain Summing ($R_f = R$): -

$Vo = -(V1 + V2 + V3)$

Observation Table: -

Sl. No.	V1 (V)	V2 (V)	R1 (kΩ)	R2 (kΩ)	Rf (kΩ)	Vo (Calculated)	Vo (Measured)
1	1.0	2	10KΩ	10KΩ	10KΩ	-3	-2.95
2	2.5	1	10KΩ	10KΩ	20KΩ	-3	-3.02
3	-1.5	3	10KΩ	20KΩ	10KΩ	0	0.02

Date

Expt. No. 08

Page No.

Title :- Simulation of Summing Amplifier using OP-Amp (Using Tinkercad / Proteus).

Theory :- An inverting Summing amplifier produces an output voltage that is proportional to the negative algebraic sum of its input voltages. It utilizes an operational amplifier configured with multiple input resistors connected to the inverting terminal, which acts as a virtual ground.

The general formula governing its operation for two input is $V_0 = -R_f \times [(V_1/R_1) + (V_2/R_2)]$. If all resistors are same value then the circuit acts as a unity gain amplifier where $V_0 = -(V_1 + V_2)$.

Component Required :-

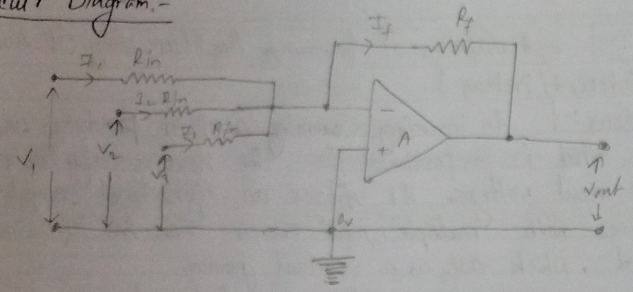
		Quantity
1.	OP-Amp (IC 741)	1
2.	Resistor (10KΩ, 20KΩ, 47KΩ)	3
3.	Multimeter (Digital Display)	1
4.	Power Supply (15V)	3
5.	Bread board	1
6.	Wires	required

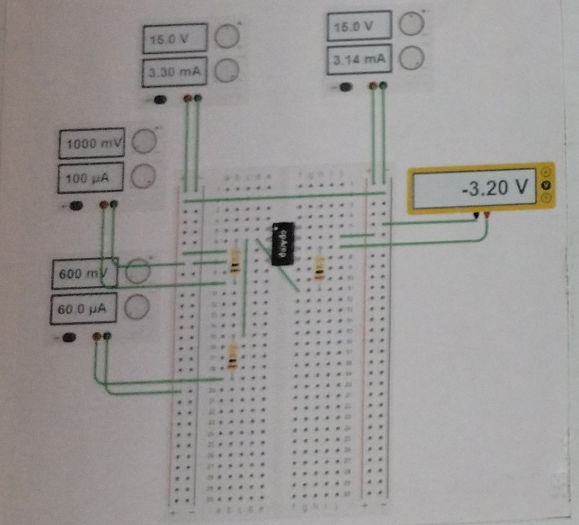
Observation Table :-

SN	V_1 (V)	V_2 (V)	R_1 (KΩ)	R_2 (KΩ)	R_f (KΩ)	V_0 (Calculated)	V_0 (Measured)
1	1.0	2	10 KΩ	10 KΩ	10 KΩ	-3	-2.95
2	0.5	1	10 KΩ	10 KΩ	10 KΩ	-3	-3.02
3	-1.5	3	10 KΩ	10 KΩ	10 KΩ	0	0.02
4	0.5	1.5	10 KΩ	10 KΩ	10 KΩ	-9.9	-9.35

Teacher's Signature

Circuit Diagram:-





Date

Expt. No. Page No.

Calculations:-

$V_1 = 1.0V$, $V_2 = 2V$, $R_1 = 10K\Omega$, $R_2 = 10K\Omega$

$R_f = 10K\Omega$, V_0 (Measured) $= -2.95V$

V_0 (Calculated) $= -R_f \times [(V_1/R_1) + (V_2/R_2)]$ V_0 (calculated)

$= -10 \times [(1.0/10) + (2/10)]$

$= -3V$

% Error $= \frac{V_0(Cal) - V_0(Mea)}{V_0(Cal)} \times 100\% = \frac{-3 - (-2.95)}{-3} = 1.67\%$

Conclusion:-

The Simulation Succesfully demonstrates the operation of an inverting Summing amplifier using an IC741 Op amp. The measured output voltages closely match the theoretically calculated algebraic sums across various resistor and voltage configurations. The minor discrepancies between the calculated and measured values validating the accuracy of the theoretical Summing amplifier formula under practical Simulation Conditions.

Teacher's Signature